INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS553)

LAB 4

Assignment 4: Selection Sort

Assignment aim: Sort arrays by selecting the minimum from the unsorted part.

Experiment No. 4.1 — Find index of smallest value

1. Experiment Title: Find index of smallest value

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: Selection sort finds the smallest element in the remaining unsorted region and swaps it into place.

5. Algorithm / Procedure:
Set the minimum index to zero; compare later elements with the current minimum; record the index of a smaller element; print the index.
Then compile and execute the Java program.

6. Program:

```
public class Task_01_Find_index_of_smallest_value  {
    public static void main(String[] args) {
        int[] a = {64, 25, 12, 22, 11};
        int minIndex = 0;
        for (int i = 1; i < a.length; i++) {
            if (a[i] < a[minIndex]) minIndex = i;
        }
        System.out.println(minIndex);
    }
}
```

7. Input: Fixed input values are declared in the program: the array {64, 25, 12, 22, 11}.

8. Output:
4

Time / Space Complexity: O(n) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What does an index represent?
- Why store minIndex rather than only the minimum value?
- What is the index of 11?

Experiment No. 4.2 — Sort an array using selection sort

1. Experiment Title: Sort an array using selection sort

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: Selection sort finds the smallest element in the remaining unsorted region and swaps it into place.

5. Algorithm / Procedure:
For each position, find the smallest value in the unsorted suffix; swap it into the current position; continue to the end.
Then compile and execute the Java program.

6. Program:

```
public class Task_02_Sort_an_array_using_selection_sort  {
    public static void main(String[] args) {
        int[] a = {64, 25, 12, 22, 11};
        for (int i = 0; i < a.length - 1; i++) {
            int minIndex = i;
            for (int j = i + 1; j < a.length; j++) {
                if (a[j] < a[minIndex]) minIndex = j;
            }
            int temp = a[i]; a[i] = a[minIndex]; a[minIndex] = temp;
        }
        for (int value : a) System.out.println(value);
    }
}
```

7. Input: Fixed input values are declared in the program: the array {64, 25, 12, 22, 11}.

8. Output:
11
12
22
25
64

Time / Space Complexity: O(n²) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What is the sorted and unsorted part?
- How does selection sort choose each next value?
- Is selection sort in-place?

Experiment No. 4.3 — Perform first selection-sort step

1. Experiment Title: Perform first selection-sort step

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: Selection sort finds the smallest element in the remaining unsorted region and swaps it into place.

5. Algorithm / Procedure:
Find the minimum element in the entire array; swap it with the first element; print the array after this first step.
Then compile and execute the Java program.

6. Program:

```
public class Task_03_Perform_first_selection{
    public static void main(String[] args) {
        int[] a = {64, 25, 12, 22, 11};
        int minIndex = 0;
        for (int i = 1; i < a.length; i++) {
            if (a[i] < a[minIndex]) minIndex = i;
        }
        int temp = a[0]; a[0] = a[minIndex]; a[minIndex] = temp;
        for (int value : a) System.out.println(value);
    }
}
```

7. Input: Fixed input values are declared in the program: the array {64, 25, 12, 22, 11}.

8. Output:
11
25
12
22
64

Time / Space Complexity: O(n) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What is the first step of selection sort?
- Which values are swapped here?
- Is the array fully sorted after one step?